

CENG381 Stochastic Processes

Recurrence Relations — Practice 1 (Answer Key)

Q1 Solution

a) Characteristic equation and roots:

```
Substitute  $a_n = r^n$ :  
 $r^2 = 5r - 6$   
 $r^2 - 5r + 6 = 0$   
 $(r - 2)(r - 3) = 0$   
 $r_1 = 2, \quad r_2 = 3 \quad (\text{distinct roots})$ 
```

b) General solution and constants:

```
 $a_n = \alpha_1 * 2^n + \alpha_2 * 3^n$   
  
 $a_0 = 2: \quad \alpha_1 + \alpha_2 = 2 \quad \dots (I)$   
 $a_1 = 5: \quad 2\alpha_1 + 3\alpha_2 = 5 \quad \dots (II)$   
  
 $(II) - 2*(I): \quad \alpha_2 = 5 - 4 = 1$   
 $(I): \quad \alpha_1 = 2 - 1 = 1$ 
```

c) Closed form, a_n , and growth rate:

```
 $a_n = 2^n + 3^n$   
  
 $a_4 = 2^4 + 3^4 = 16 + 81 = 97$   
  
As  $n \rightarrow \infty$ :  $3^n$  dominates (since  $3 > 2$ ).  
 $a_n \sim 3^n$ , i.e. the sequence grows exponentially at rate 3.
```

Q2 Solution

a) Characteristic equation and repeated root:

```
 $r^2 - 6r + 9 = 0$   
 $(r - 3)^2 = 0$   
 $r = 3 \quad (\text{double root})$ 
```

b) General solution for repeated root:

```
 $a_n = (\alpha_1 + \alpha_2 * n) * 3^n$ 
```

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$$\begin{aligned}a_0 = 1: & \quad \alpha_1 * 1 = 1 \Rightarrow \alpha_1 = 1 \\a_1 = 6: & \quad (1 + \alpha_2) * 3 = 6 \Rightarrow 1 + \alpha_2 = 2 \Rightarrow \alpha_2 = 1\end{aligned}$$

c) Closed form and comparison:

$$a_n = (1 + n) * 3^n$$

$$a_3 = (1 + 3) * 3^3 = 4 * 27 = 108$$

Growth rate: both Q1 and Q2 grow at rate 3^n .

The extra factor n makes the repeated-root solution grow slightly

faster ($n*3^n$ vs 3^n), but the exponential base is the same.

Q3 Solution

a) Homogeneous part:

$$\begin{aligned}a_n &= 4 * a_{(n-1)} \Rightarrow r - 4 = 0 \Rightarrow r = 4 \\a_n^{(h)} &= \alpha_1 * 4^n\end{aligned}$$

b) Particular solution via $cn + d$:

Substitute $a_n = cn + d$ into the full recurrence:

$$\begin{aligned}cn + d &= 4*[c(n-1) + d] + 3n \\cn + d &= 4cn - 4c + 4d + 3n\end{aligned}$$

$$\text{Collect coefficient of } n: \quad c = 4c + 3 \Rightarrow -3c = 3 \Rightarrow c = -1$$

$$\begin{aligned}\text{Collect constants:} \quad d &= -4c + 4d \Rightarrow -3d = -4c = 4 \Rightarrow \\d &= -4/3\end{aligned}$$

$$\text{Particular solution: } a_n^{(p)} = -n - 4/3$$

c) Full solution and verification:

$$a_n = \alpha_1 * 4^n - n - 4/3$$

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Apply $a_1 = 2$:

$$4\alpha_1 - 1 - 4/3 = 2$$

$$4\alpha_1 = 2 + 1 + 4/3 = 3 + 4/3 = 13/3$$

$$\alpha_1 = 13/12$$

Closed form: $a_n = (13/12)4^n - n - 4/3$

Verify a_2 from the recurrence:

$$a_2 = 4a_1 + 3 \cdot 2 = 4 \cdot 2 + 6 = 14$$

$$\text{From formula: } a_2 = (13/12)16 - 2 - 4/3$$

$$= 208/12 - 24/12 - 16/12 = 168/12 = 14 \quad \checkmark$$